

The Extraspecial Layer of Barnes–Wall Symmetry

Proof of the four-sector sigma–MGF and direct matrix verification

Computational verification note

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Abstract

For the plus-type extraspecial real Pauli group $E_m \cong 2_+^{1+2m}$ on $V_m = \mathbb{C}^{2^m}$, we derive the complete four-sector determinant average. An independent program constructs every matrix $\pm X^a Z^b$ and checks 24 one-row and mixed coefficients in dimensions 2, 4, 8. All comparisons are exact integers and all pass.

1 Representation and coefficient target

Put $N = 2^m$. On the basis $\{e_x : x \in \mathbb{F}_2^m\}$ define

$$X(a)e_x = e_{x+a}, \quad Z(b)e_x = (-1)^{b \cdot x} e_x.$$

Then

$$E_m = \{\pm X(a)Z(b) : a, b \in \mathbb{F}_2^m\}, \quad |E_m| = 2N^2.$$

For a partition $\tau = (\tau_1, \dots, \tau_r)$, determinant expansion gives

$$[m_\tau] \mathfrak{M}_{E_m, m} = \frac{1}{|E_m|} \sum_{g \in E_m} \prod_i h_{\tau_i}(\text{spec } g), \quad (1)$$

where $\sum_{d \geq 0} h_d(\text{spec } g) t^d = \det(I - tg)^{-1}$. This is exactly $\dim(\bigotimes_i \text{Sym}^{\tau_i} V_m)^{E_m}$.

2 The four spectral sectors

The central elements $I, -I$ have determinants $(1 - t)^N, (1 + t)^N$. For $g = \pm X(a)Z(b)$ one has

$$g^2 = (-1)^{a \cdot b} I.$$

Off the center, the trace is zero. Hence an involution has $N/2$ eigenvalues of each sign, while an element squaring to $-I$ has $N/2$ eigenvalues each of i and $-i$. Their determinants are respectively $(1 - t^2)^{N/2}$ and $(1 + t^2)^{N/2}$.

The quadratic form $q(a, b) = a \cdot b$ is plus type. It has $(N^2 + N)/2$ zeros, including $(0, 0)$. Accounting for both central lifts therefore gives

$$\#\{g \notin Z(E_m) : g^2 = I\} = N^2 + N - 2, \quad \#\{g : g^2 = -I\} = N^2 - N.$$

The counts sum with the two central elements to $2N^2$. Averaging proves

$$\mathfrak{M}_{E_m, m}(\mathbf{t}) = \frac{1}{2N^2} \left[\prod_j (1 - t_j)^{-N} + \prod_j (1 + t_j)^{-N} \right. \\ \left. + (N^2 + N - 2) \prod_j (1 - t_j^2)^{-N/2} \right. \\ \left. + (N^2 - N) \prod_j (1 + t_j^2)^{-N/2} \right]. \quad (2)$$

Because $-I \in E_m$, the first two terms cancel in odd total degree and the other two contain only even powers. Thus every odd-total-degree coefficient vanishes.

3 Independent coefficient extraction

The verifier does not classify matrices after construction. For each explicit matrix it computes the eigenvalues, obtains h_d recursively from the product generating function, forms the integrand in (1), and averages. The formula side separately extracts coefficients from the four factors in (2) using binomial coefficients. Before coefficient extraction, every matrix is independently checked to have real entries and to satisfy $g^\top g = I$.

m	N	$ E_m $	τ	direct	formula
1	2	8	(3, 1)	2	2
1	2	8	(2, 2)	3	3
2	4	32	(4)	5	5
2	4	32	(4, 2)	23	23
3	8	128	(3, 1)	15	15
3	8	128	(6)	29	29
3	8	128	(4, 2)	190	190

The full sweep uses eight partitions at each rank, including the odd case (1) and the mixed cases (1, 1), (3, 1), (2, 2), (4, 2). All 24 pass.

4 Reproduction

```
.venv/bin/python for_this_guy/barnes_wall_lattice_automorphism_groups/
  extraspecial_real_pauli/verification.py
.venv/bin/marimo edit for_this_guy/barnes_wall_lattice_automorphism_groups/
  extraspecial_real_pauli/notebook.py
```

Join each wrapped command onto one shell line. The computation is exhaustive at the stated ranks; no random sampling or precomputed character table is used.